

RESEARCH ARTICLE

Peripheral blood mononuclear cell secretome from patients with autoimmune encephalitis promotes seizures in vitro

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Abstract

Objective: Autoimmune encephalitis (AE) is characterized by inflammatory processes in the central nervous system and frequently presents with seizures. Even though an ictogenic potential has been shown for some antibodies against neuronal surface antigens (NSAbs), AE pathophysiology is complex, and NSAbs-independent mechanisms are likely to contribute to seizures. We investigated whether the secretome released by peripheral blood mononuclear cells (PBMCs) from AE patients contributes to seizure generation independently of NSAbs.

Methods: PBMCs were isolated from 19 patients with AE (including both those with and those without detectable NSAbs) and 13 healthy volunteers. After 6 h in culture, the PBMC supernatant (secretome) was infused into a heterologous in vitro whole guinea pig brain preparation. Neurophysiological activity was monitored in the isolated in vitro guinea pig brain preparation during coperfusion of PBMC-derived supernatant (secretome) with the endotoxin lipopolysaccharide and human recombinant serum albumin, to induce and mimic, respectively, a mild functional blood–brain barrier impairment. Morphological analysis of ionized calcium-binding adapter molecule 1-positive glial cells was performed in these brains after the electrophysiological experiment. Secretome obtained after 6 h in culture was analyzed with the Multiplex ELLA array system for inflammatory mediator detection.

Results: Electrophysiological recordings and immunofluorescence analyses revealed that the secretome from PBMCs derived from AE patients induced seizure-like events and microglial activation in our in vitro brain preparation. Multiplex ELLA immunoassay analysis showed significantly lower concentration of interleukin (IL)-10, IL-1Ra, tumor necrosis factor- α , IL-2, and IL-6 in the secretome of

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PBMCs derived from AE patients compared to healthy subjects. IL-1 β levels were comparable in the secretome of PBMCs derived from AE and healthy subjects.

Significance: Our findings suggest, for the first time, that peripheral inflammatory mediators could represent a trigger factor for seizure activity in AEs, beyond a possible antibody-mediated mechanism.

KEYWORDS

autoimmune encephalitis, PBMC, seizure

1 | INTRODUCTION

Autoimmune encephalitis (AE) is a group of inflammatory disorders of the central nervous system often associated with the presence of autoantibodies against either intracellular or surface neuronal proteins (NIAbs and NSAbs). Seizures are a key feature of AE, occurring in 33%–100% of cases.¹ Whereas in the AE associated with NIAbs the chronic predisposition to seizures is ascribed to residual damage resulting from cytotoxic T cell-induced neuronal death, in those associated to NSAbs, a direct effect of the antibodies has been suggested by *in vitro* and *in vivo* experiments using leucine-rich glioma-inactivated 1 protein (LGI1)² antibodies and glutamate and γ -aminobutyric acid type A (GABA_A) receptors (anti-N-methyl-D-aspartate receptor [NMDAR],^{3,4} anti- α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor [AMPA],⁵ and anti-GABA_A receptor⁶) antibodies. Despite the prominent role of NSAbs in AE pathogenesis, seizures do not invariably occur in all patients with AE, even in those sharing the same NSAb specificity. Conversely, seizures can occur in patients with clinically defined AE without detectable NSAbs. These data suggest that other NSAb-independent mechanisms could be at play in AE pathogenesis to account for seizure generation. Inflammatory mediators released by both resident and circulating immune cells contribute to increase brain excitability and/or decrease seizure threshold. In turn, seizures *per se* are able to induce inflammatory mediators released by both resident and circulating immune cells.^{7,8} In line with this hypothesis, several agents have been recently used in AE to selectively modulate the humoral and/or cellular immune system with therapeutic purposes.⁷

We aimed to verify whether the secretome produced by circulating peripheral blood mononuclear cells (PBMCs) in AE patients could contribute to seizure generation. Therefore, we decided to study the role of the secretome from PBMCs derived from AE patients in sustaining seizure activity in the *in vitro* guinea pig brain, a preparation that retains the physiological interactions between neuronal, glial, and vascular compartments in a close to *in vivo*

Key points

- Despite the prominent role of NSAbs in AE pathogenesis, seizures do not invariably occur in all patients with AE, even in those sharing the same NSAb specificity.
- Mechanisms independent of NSAbs may contribute to the pathogenesis of AE, possibly accounting for seizure generation.
- The identification of novel soluble biomarkers that could be potentially targeted in the future is required to develop new therapies for patients with AE.

condition.⁸ It represents an ideal preparation to study the specific direct effects of blood-derived molecules on brain inflammation and excitability in an intact, undamaged brain; the absence of peripheral immune cells or blood-borne molecules⁹ makes this preparation ideal to test the capability of the secretome from PBMCs derived from AE patients to sustain seizure activity. We hypothesized a novel mechanism of AE seizure generation that has the potential to develop novel therapeutic strategies to counteract seizure occurrence in this immune-mediate seizure disorder.¹⁰

2 | MATERIALS AND METHODS

Experimental procedures were carried out in compliance with the international guidelines and laws (2010/63/EU Legislation). The experimental protocols were approved by the ethics committee of the Fondazione IRCCS Istituto Neurologico Carlo Besta and by the Italian Ministry of Health (Aut Min No. 271-2012/A). Patients and healthy volunteers were selected at the epilepsy units of the institutes involved, and all subjects signed an informed consent in accordance with the Declaration of Helsinki and with approval from the ethics committees.

2.1 | Patient and healthy volunteer recruitment

Healthy volunteers ($n=13$) and patients with AE ($n=19$) were selected according to Graus criteria¹¹ at onset or in the active phase of the disease. Briefly, inclusion criteria for patients were: (1) age ≥ 18 years, (2) clinical onset no more than 9 months and at least one seizure in the month before blood sampling, (3) Antibody Prevalence in Epilepsy and Encephalopathy score ≥ 4 ,¹² and (4) no use of immunomodulating drugs in the past 3 months. All patients underwent a complete clinical characterization to assess neurological and cognitive status changes, psychiatric symptoms, disability level (modified Rankin Scale¹³), autonomic dysfunction, motor dyskinesias, and seizure semiology. Viral prodromes and systemic cancer history were specifically searched for. All patients underwent cerebrospinal fluid (CSF) analysis for signs of inflammation (elevated proteins, pleocytosis, oligoclonal band). Specific antibodies of autoimmune encephalitis (against NMDAR, AMPAR, GABA receptor, contactin-associated protein 2 (CASPR2), LGI1, glutamic acid decarboxylase 65, dipeptidyl-peptidase-like protein 6 (DPPX), IgLon5, Hu, Yo, Ri, Ma1, and Ma2/Ta) were investigated in both CSF and serum.

The identification of serum and CSF neuronal surface autoantibodies was performed locally at each center using a commercial fixed cell-based assay (CBA) panel including NMDAR, LGI1, CASPR2, GABAB, AMPAR, and DPPX (Euroimmun). In addition, results were confirmed using in-house tissue-based assay and live CBA when necessary, according to the clinical suspicion.

Electroencephalography (EEG), neuropsychological testing, brain magnetic resonance imaging, and brain fluorodeoxyglucose positron emission tomography were routinely performed; long-term EEG/video-EEG monitoring and study of the vegetative system were performed when necessary. Healthy volunteers without a history of the neurological or immunological disease were recruited as control subjects.

Patients' clinical features and treatment history are summarized in Table 1.

2.2 | PBMC isolation

Human PBMCs (hPBMCs) were isolated from anticoagulated whole blood using a discontinuous density gradient (Lymphoprep; Nycomed), as previously described (Figure 1).¹⁴ hPBMCs were treated with $1 \times \text{RBC}$ Lysis Buffer (eBioscience) for red blood cell lysis, according to the manufacturer protocol. Immune cells were cultured in 24-well plates (time 0 [t0]) at a density of 2×10^6 cells/well in 1 mL RPMI 1640 (EuroClone) supplemented with L-glutamine ($2 \text{ mmol} \cdot \text{L}^{-1}$), sodium pyruvate ($1 \text{ mmol} \cdot \text{L}^{-1}$),

nonessential amino acids ($.1 \text{ mmol} \cdot \text{L}^{-1}$), penicillin ($100 \text{ U} \cdot \text{mL}^{-1}$), streptomycin ($.1 \text{ mg} \cdot \text{mL}^{-1}$), hydroxyethylpiperazine ethane sulfonic acid (HEPES) buffer ($.01 \text{ mol} \cdot \text{L}^{-1}$), and 10% fetal bovine serum.

2.3 | Cell viability assay

Cell viability was assessed at t0 and t6 using the Trypan blue method. Briefly, cells in their culture medium were added to a Trypan blue solution (Sigma-Aldrich) at a 1:10 ratio. Cells were counted using a hemocytometer under a light microscope.

2.4 | Isolated guinea pig brain preparation and electrophysiological recordings

Brains were isolated from young adult (3–4 weeks old) Hartley guinea pigs ($n=22$, 150–200g, Charles River Laboratories) according to the standard technique described elsewhere (Figure 1).⁸ Briefly, after intraperitoneal barbiturate anesthesia ($125 \text{ mg} \cdot \text{kg}^{-1}$), guinea pigs were transcardially perfused with a cold (15°C) oxygenated ($95\% \text{ O}_2/5\% \text{ CO}_2$) solution composed of $126 \text{ mmol} \cdot \text{L}^{-1} \text{ NaCl}$, $3 \text{ mmol} \cdot \text{L}^{-1} \text{ KCl}$, $1.2 \text{ mmol} \cdot \text{L}^{-1} \text{ KH}_2\text{PO}_4$, $1.3 \text{ mmol} \cdot \text{L}^{-1} \text{ MgSO}_4$, $2.4 \text{ mmol} \cdot \text{L}^{-1} \text{ CaCl}_2$, $26 \text{ mmol} \cdot \text{L}^{-1} \text{ NaHCO}_3$, $15 \text{ mmol} \cdot \text{L}^{-1}$ glucose, $2.1 \text{ mmol} \cdot \text{L}^{-1}$ HEPES, and 3% dextran (molecular weight = 70 000 Da, pH 7.3). Following decapitation, brains were rapidly dissected out and transferred to a recording chamber with its ventral surface upward to visualize the base of the brain and the resident vascular system. Under a stereomicroscope, the dura mater enfolding the basilar artery was gently removed, and a polyethylene cannula (terminal gauge = .25 mm) was inserted into the basilar artery. The cannula was tied to the basilar artery with a silk thread, and the brain circulation was restored by arterial perfusion with the above solution (pH 7.3) at a rate of $6 \text{ mL} \cdot \text{min}^{-1}$ via a peristaltic pump (Minipuls 4, Gilson) connected to a set of Tygon tubing. Then, the carotid and the hypophyseal arteries were ligated with silk knots to re-establish the physiological cerebral vascular flow through the Willis circle without arterial leaks. The temperature of the incubation chamber was successively increased from 15°C to 32°C (room temperature) at a rate of $.2^\circ\text{C} \cdot \text{min}^{-1}$, with a controlled ramp to heat exchangers (Peltier Devices) commanded by a thermostat connected to a temperature controller (RG-01, NPI Electronic).

Extracellular recordings were performed in the piriform cortex (PC), in the medial entorhinal cortex (mEC), and in the hippocampal formation dentate gyrus (DG) region of the isolated brains using .9% NaCl-filled glass electrodes obtained with a pipette puller (Harvard Apparatus)

TABLE 1 Clinical features and treatment history of patients and HC.

Etiology	Anti-NMDAR, <i>n</i> = 2	Anti-CASPR2, <i>n</i> = 4	Anti-LGII, <i>n</i> = 6 ^a	Negative NSAbs, <i>n</i> = 7	Whole cohort	HC, <i>n</i> = 13
Gender, F/M	0/2	0/4	2/4	2/5	19	7/6
Age, years, median (range)	45.5 (44–47)	69.5 (67–70)	70.5 (59–82)	58 (43–70)	62 (43–82)	51 (25–66)
Disease duration, median (range) ^b	1 (1)	9 (1–9)	2.5 (1–9)	8 (2–9)	8 (1–9)	—
Seizure frequency, median (range) ^c	1 (1)	1 (1)	20.5 (1–450)	8 (1–180)	8 (1–450)	—
Immunotherapy ^d	No	No	No	No	No	No
AEDs, <i>n</i> ^e	0	1 (gabapentin given for neuropathic pain)	0	1 or 2	0–2	—
Significant comorbidities	Migraine; Wolff–Parkinson–White syndrome	Prostate adenocarcinoma	Hypothyroidism; type 2 diabetes mellitus	Aortic valvulopathy, bilateral sensorineural hearing loss, essential tremor; migraine	—	—

Note: Statistical analysis was performed using chi-squared and Fisher exact test for categorical variables. Results are expressed as median (lower and upper values are reported).

Abbreviations: AED, antiepileptic drug; CASPR2, contactin-associated protein 2; F, female; HC, healthy controls; LGII, leucine-rich glioma-inactivated 1 protein; M, male; NMDAR, N-methyl-D-aspartate receptor; NSAbs, antibodies against neuronal surface antigens.

^aFacio-brachial dystonic seizure multiple times per day.

^bMonths from first clinical onset to blood collection.

^cNumber of seizure per months at blood collection.

^dImmunotherapy before blood collection.

^eAt time of sampling.

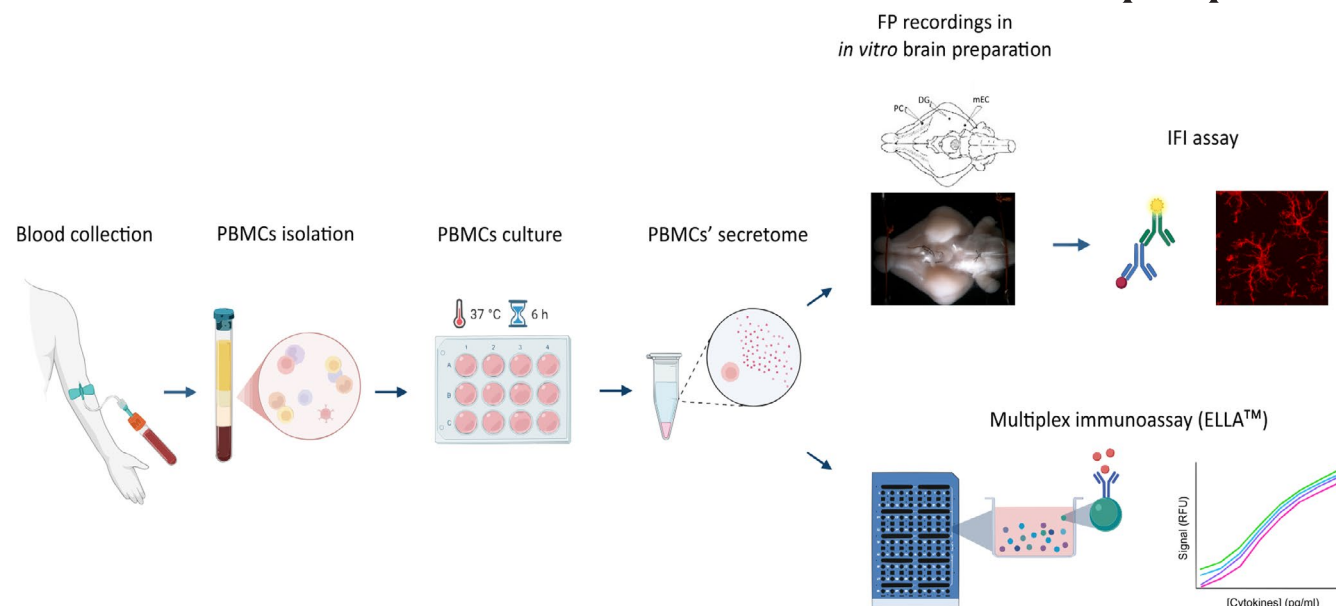


FIGURE 1 Experimental design. Healthy volunteers and selected autoimmune encephalitis (AE) or suspected AE patients were recruited and whole blood samples were collected. Then, peripheral blood mononuclear cells (PBMCs) were isolated from whole blood and put in culture. After 6 h, PBMC secretome was collected, and it was infused in the *in vitro* brain preparation. At the end of each experiment, brains were processed for immunofluorescence analysis. PBMC secretome was analyzed by Multiplex immunoassay ELLA. Created in <https://BioRender.com>. DG, dentate gyrus; FP, Field Potential; IFI, Indirect Immunofluorescence; mEC, medial entorhinal cortex; PC, piriform cortex; RFU, relative fluorescence units.

with 2–5 MΩ resistance, positioned under stereoscopic microscope (Nikon) control. Responses evoked by electrical stimulation of the lateral olfactory tract through an isolation unit driven by a stimulus isolation unit (SIU5, Grass Instruments Division, Astro-Med) were used to monitor the viability of the preparation during the experiment. Field potential signals were amplified with a multichannel differential amplifier (DBA8-v6, Supertech Instruments), high-pass filtered at 1 or .05 Hz, acquired with DIGIDATA 1440A (Axon CNS, Molecular Devices; Clampex 11), and analyzed with Clampfit (v11.2).

Supernatant (5 mL) derived from hPBMCs after 6 h *in vitro* incubation without any stimulation was collected for the experiments on the isolated guinea pig brain, diluted to a final volume of 10 mL with phosphate-buffered saline (PBS) without calcium and magnesium, and perfused into the isolated brain preparation for 30 min at the rate of .3 mL/min via a syringe-infuser (KD Scientific, Harvard Apparatus) connected to the brain main arterial perfusion line.¹⁵

2.5 | Morphological analysis of microglial cells

At the end of each experiment, isolated brains were post-fixed by immersion into 4% paraformaldehyde in PBS

(.1 mol·L⁻¹, pH 7.4) and were processed for immunofluorescence analysis (Figure 1).

A standardized protocol was used for immunohistochemical staining. After nonspecific antigen binding sites were blocked with 1% bovine serum albumin (BSA; BSA/10% normal donkey serum [NDS]/.2% Triton-X 100 in PBS), free-floating sections were incubated overnight at 4°C with ionized calcium-binding adapter molecule 1 (Iba1; 1:1000, Abcam) in .1% BSA/1% NDS. On the subsequent day, sections were incubated for 75 min in the secondary antibodies conjugated with Alexa Fluor 555 (1:600, Thermo Fisher Scientific) diluted in .1% BSA/1% NDS and counterstained with 4,6-diamidino-2-phenylindole (DAPI; 1:1000, Thermo Fisher Scientific). Tissue was washed in PBS and then coverslipped with Fluoromount (Calbiochem). For morphological evaluation of microglial cells, two immunostained coronal sections per animal were visualized using an LSM980 Confocal with Airyscan (Carl Zeiss). Previews of the whole section in wide field (10×/.3 dry) using the DAPI channel were taken to choose the areas of interest in the ventral CA1 stratum radiatum, which was further acquired at a higher resolution with the confocal mode. Two channel (Iba1 and DAPI) Z-stack images (Z-step intervals of .2 μm) were acquired with a 63×/1.4 oil objective, using excitation of 358 nm and 555 nm (Laser Ar), and with a x-y sampling of 72 nm. Laser intensity was set at 30%–35% power, and gain and

photomultiplier were kept constant during acquisition of all images. Cells were selected for morphological analysis if the following criteria were met: (1) the colocalization of the Iba1-positive cell with a single DAPI-stained nucleus; (2) the absence of truncated processes; and (3) a sufficient distance from neighboring cells to ensure correct reconstruction of single cells. A total of 59 cells (five regions of interest per animal) were selected from isolated brain preparations treated with anti-NMDAR ($n=1$, $n=5$ cells), anti-CASPR2 ($n=2$, $n=10$ cells), and anti-LGI1 ($n=2$, $n=10$ cells), without detectable NSAbs ($n=3$, $n=15$ cells) in patient-derived hPBMC secretome and from isolated brains treated with healthy control-derived hPBMCs ($n=4$ subjects/brains; $n=19$ cells). Morphological analysis was performed by calculating the ramification index (RI), defined as the ratio between the cell area (Ac) and the convex hull area (CHA), using Fiji ImageJ software (v2.14.0).¹⁶ CHA represents the smallest convex polygon containing the whole cell shape, whereas Ac is defined as the area covered by the cell. Resting microglial cells are characterized by small cell bodies and long ramified processes, resulting in small Ac and large CHA, whereas activated microglia show more similar Ac and CHA due to hypertrophy of somata and processes, resulting in RI values closer to 1 for activated microglia and RI values closer to 0 for resting microglia.¹⁷

2.6 | Multiplex ELLA immunoassay

Supernatants derived from hPBMC cultures after 6 h of in vitro incubation without any stimulation were tested with the Multiplex ELLA immunoassay (Bio-Techne), according to the manufacturer protocol (Figure 1).

2.7 | Statistical analysis

The normal distribution of samples was checked with the Shapiro–Wilk test, and the homogeneity of variances was evaluated with the *F*-test. When the equal variance criterion was violated, the Welch correction was used. The nonparametric Mann–Whitney test was chosen when data were not normally distributed. Otherwise, Student *t*-test was used. For Gaussian distributed data, mean $\pm$ SD and *p*-value are reported. For nonnormally distributed data, median value and *p*-value are reported. All statistical tests were performed with Origin 9.0 (OriginLab Corporation). The tests were two-sided, and a confidence interval of 95% ($p \leq .05$) was required for values to be statistically significant. The data illustrated with boxplots show median (central lines), quartiles (25% and 75%; boxes), and ± 1.5 times the interquartile range (whiskers).

3 | RESULTS

The selective effect of PBMC secretome obtained from 13 healthy volunteers and 19 patients with AE either positive for antibodies against NMDAR, LGI1, and CASPR2 ($n=2$, 6, and 4, respectively) or autoantibody-negative ($n=7$) was investigated by perfusing hPBMC secretome in the in vitro brain. The endotoxin lipopolysaccharide (LPS; 100 ng/mL, Sigma-Aldrich) and human recombinant albumin (hALB; 4 g/L, Sigma-Aldrich) were injected into the arterial stream before performing hPBMC secretome perfusion to, respectively, induce and mimic a mild condition of blood–brain barrier (BBB) impairment (Figure 2A, protocols a and b).

Simultaneous extracellular recordings performed in the PC (data not shown), mEC, and DG of the hippocampus confirmed our laboratory's previous findings that treatment with LPS and hALB ($n=3$; data not shown) does not induce any electrophysiological alterations.¹⁵ Neither did arterial perfusion with LPS + hALB + hPBMC secretome derived from healthy subjects (t_6 , 10×10^6 cells, $n=13$, data not shown) ever evoke any epileptiform activity.

In contrast, treatment with LPS + hALB + hPBMC secretome derived from patients with AE (t_6 , 10×10^6 cells, $n=19$; Figure 2B,C) evoked seizurelike events (SLEs) in mEC and DG. SLEs were defined as no physiological events longer than 10 s characterized by fast activity at onset that evolves into highly synchronous burst/spike discharges. These results reinforce the concept that PBMC secretome derived from patients with AE is an essential requirement for the increase in excitability.^{18,19} Interestingly, extralimbic brain regions, such as the PC, did not generate epileptiform activities.

To assess the effects of hPBMC secretome on microglial functional status, the RI of these cells was evaluated in brain slices from both experimental protocols (Figure 3). The RI, calculated by the ratio of the Ac (middle row in Figure 3A) divided by the CHA (bottom row in Figure 3A), showed values closer to 0 for LPS + hALB + healthy volunteer hPBMC secretome treatment (Figure 3B, blue column; $.16 \pm .02$ vs. $.39 \pm .09$) and values closer to 1 for LPS + hALB + AE patient hPBMC secretome treatment (Figure 3B, red column; $p \leq .001$). Microglial cells treated with healthy volunteer hPBMC secretome displayed a resting ramified morphology, with a high number of thin processes (left column, Figure 3A). Conversely, microglia from brains treated with AE patient-derived hPBMC secretome displayed an activated amoeboid shape (2nd to 5th columns from the left, Figure 3A).

We measured the levels of inflammatory mediators in hPBMC secretome derived from patients with AE ($n=19$) and healthy subjects ($n=13$; Figure 4). Similar mortality rates in hPBMC cultures were detected in AE patients and

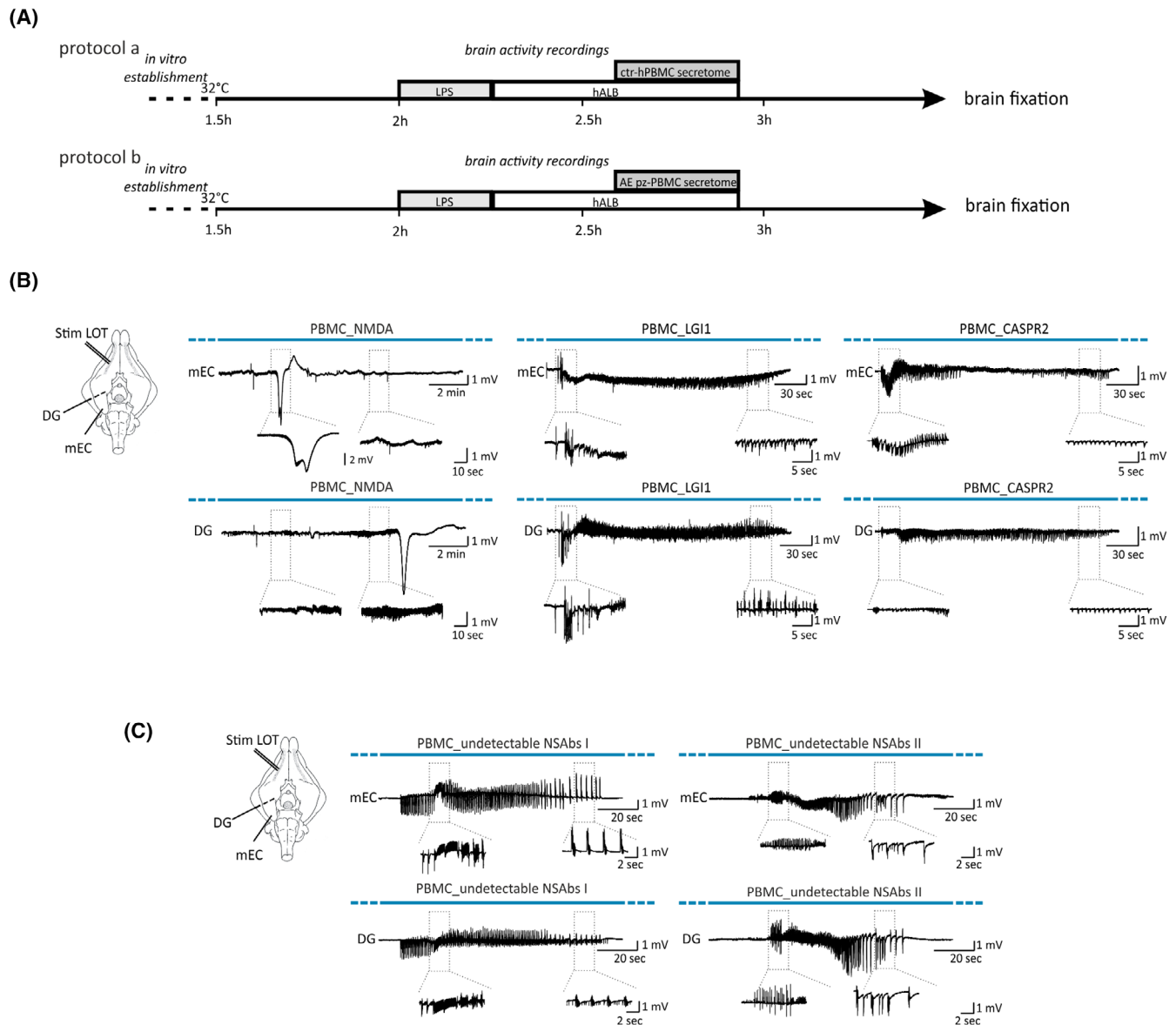


FIGURE 2 Effects of different experimental protocols on brain excitability. (A) Schematic drawing of the different experimental protocols utilized in the study. The *in vitro* brain was arterially perfused with the following: (protocol a) lipopolysaccharide (LPS), human recombinant albumin (hALB), and healthy volunteer (ctr) peripheral blood mononuclear cell (PBMC) secretome; or (protocol b) LPS, hALB, and autoimmune encephalitis (AE) or suspected AE patient-derived PBMC secretome. Each experimental protocol lasted 4 h. (B) Seizurelike events (SLEs) simultaneously recorded in medial entorhinal cortex (mEC) and in hippocampal dentate gyrus (DG) after arterial perfusion with PBMC secretome of patients positive for anti-N-methyl-D-aspartate (NMDA) receptor, anti-leucine-rich glioma-inactivated 1 protein (LGI1), and anti-contactin-associated protein 2 (CASPR2) and (C) AE patients without detectable antibodies against neuronal antigens. The application of protocol a did not induce SLEs. hPBM, human PBM; LOT, lateral olfactory tract; NSAb, antibody against neuronal surface antigens; pz, patient.

healthy subjects. Significantly lower levels of interleukin (IL)-10, IL-6, IL-2, tumor necrosis factor (TNF)- α , and IL-1ra were measured in secretome from patients with AE. IL-1 β levels were comparable in the secretome of PBMCs derived from AE and healthy subjects. AE-derived versus healthy subject-derived hPBM IL-10, IL-6, IL-2, TNF- α , IL-1ra, and IL-1 β values were respectively 1.41 versus 2.145 pg/mL ($p \leq .01$), 7.4 versus 29 pg/mL ($p \leq .05$), .065 versus .2315 pg/mL ($p \leq .05$), 3.99 versus 7.31 pg/mL

($p \leq .05$), 654 versus 1446 pg/mL ($p \leq .05$), and 1.64 versus 5.21 pg/mL ($p = .1$).

4 | DISCUSSION

Our findings suggest that multiple inflammatory mediators expressed in the secretome derived from AE PBMCs could represent a pathogenic contributor to seizures in

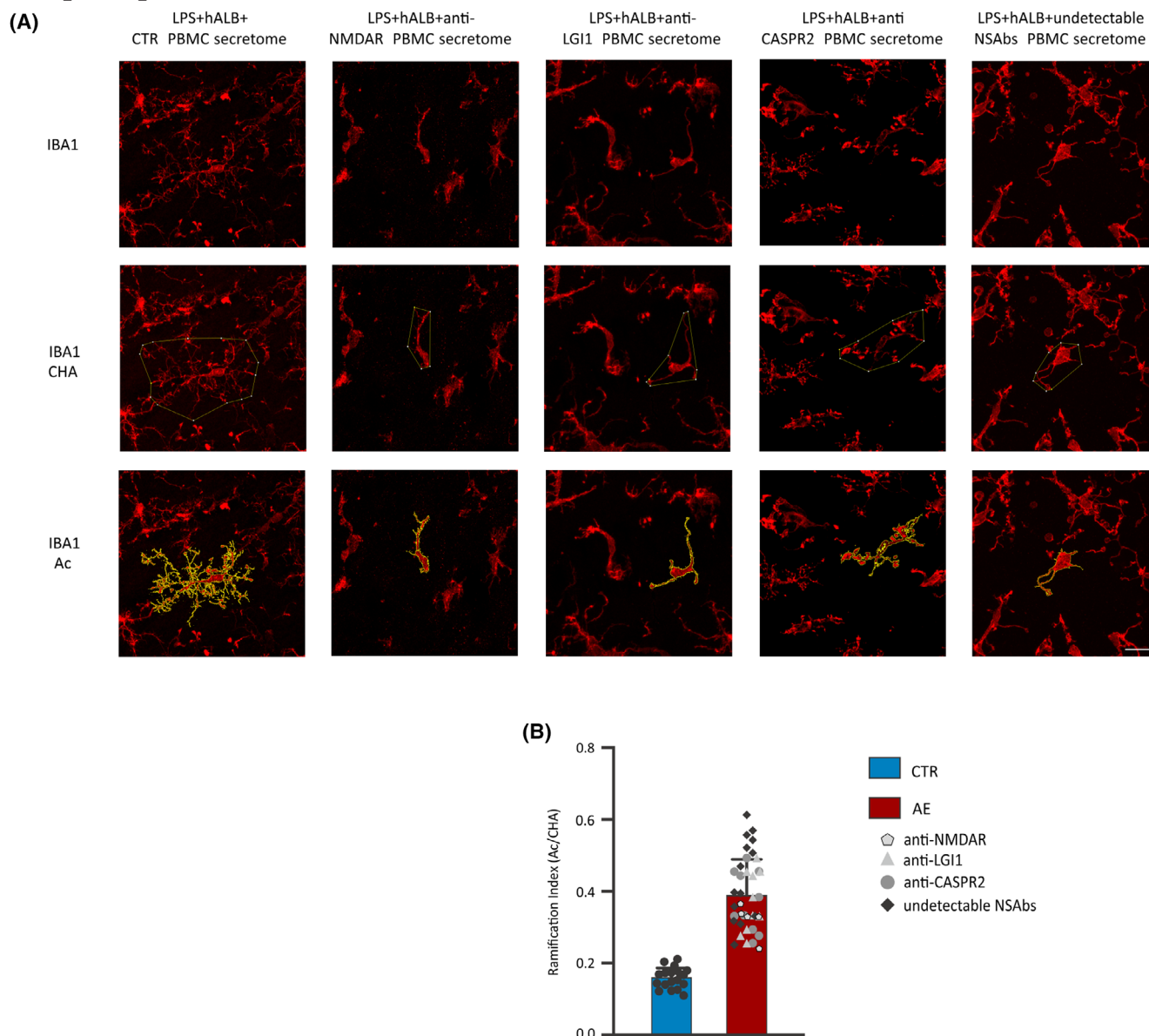


FIGURE 3 Ramification index (cell area [Ac]/convex hull area [CHA]) analysis of microglia in ventral hippocampus. (A) Representative morphologies of microglial cells in ionized calcium-binding adaptor molecule 1 (IBA1) immunofluorescence coronal sections of the ventral hippocampal region are represented for control (CTR) peripheral blood mononuclear cell (PBMC) secretome, anti-N-methyl-D-aspartate receptor (NMDAR), anti-leucine-rich glioma-inactivated 1 protein (LGI1), anti-contactin-associated protein 2 (CASPR2), and undetectable antibodies against neuronal surface antigens (NSAbs) PBMC secretome. (B) Ac and CHA were measured. Ac represents the area actually occupied by a cell and CHA is the smallest convex polygon containing the whole cell shape. Calibration bar = 20 μ m. All experiments were done on five cells per animal. AE, autoimmune encephalitis; hALB, human recombinant albumin; LPS, lipopolysaccharide.

patients with AE. Epileptic seizures not explained by a previously identified seizure disorder represent one of the criteria for the diagnosis of AE.²⁰ Clinical and experimental evidence suggests that inflammatory processes contribute to the pathophysiology of acute symptomatic seizures and epilepsy, but the underlying mechanisms remain mostly unknown.^{21,22} NSAbs have been proposed to exert a direct effect on neural function, and the removal of pathogenic NSAbs has been considered the main target of immunotherapy in these patients. Whether circulating

autoantibodies are the main pathogenic actor in ictogenesis in AE is still a matter of debate. Ictogenic properties have been demonstrated only for a subgroup of NSAbs/NIAbs, and not all of these patients develop seizures. This might suggest that other mechanisms are implicated, one of which could be systemic inflammation. In the past decade, a high prevalence of systemic autoimmune disorders has been reported among people with epilepsy, even in the absence of structural lesions or specific NSAbs,^{23,24} suggesting that systemic inflammation might

PBMC secretome at t6

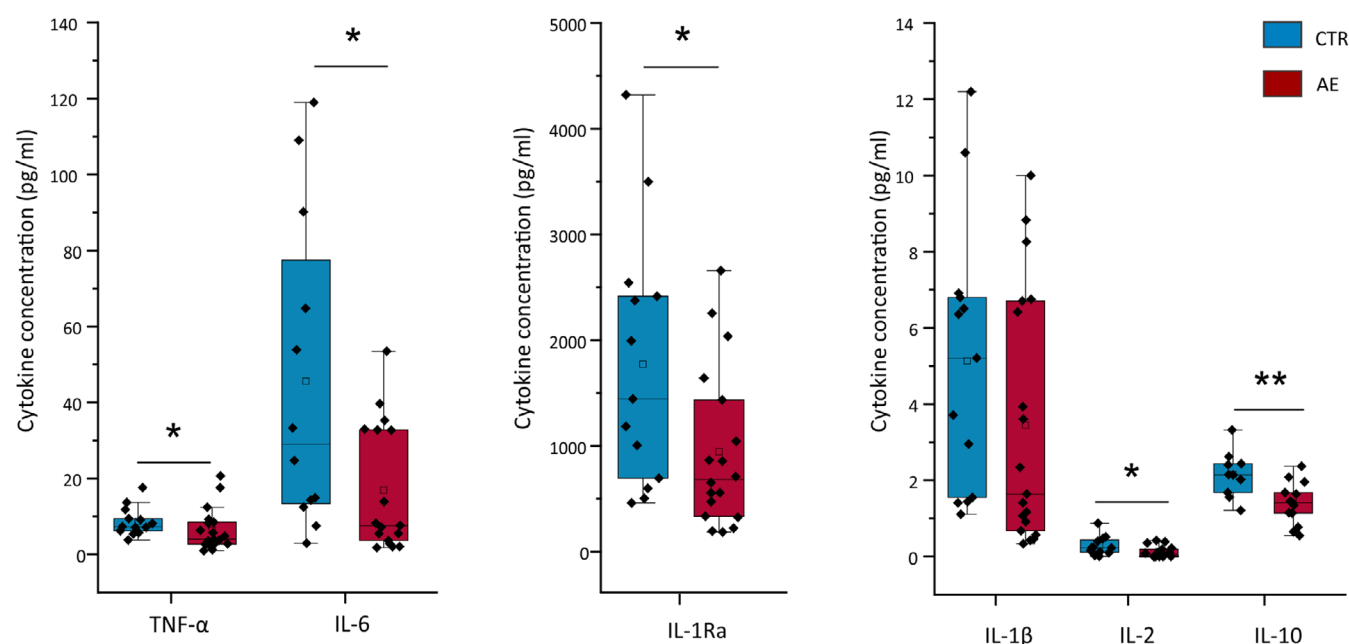


FIGURE 4 Cytokine concentration in healthy controls (CTR) and autoimmune encephalitis (AE) or suspected AE patients in peripheral blood mononuclear cell (PBMC) secretome at time 6 (t6). Concentrations of the inflammatory mediators involved in ictogenesis were evaluated with Multiplex ELLA immunoassay. * $p \leq .05$ and ** $p \leq .01$ by Mann–Whitney test. Median (central lines), quartiles (25% and 75%; boxes), and ± 1.5 times the interquartile range (whiskers) are represented. IL, interleukin; TNF, tumor necrosis factor.

lead to neuroinflammation and directly contribute to seizures. The innate arm of immunity can also be involved in epileptogenesis by increased synthesis and release of proinflammatory cytokines and chemokines.²² These systemic mediators can activate BBB cellular components, leading to the extravasation of serum albumin, peripheral immune cells, and inflammatory molecules in the brain, where they can collectively enhance neuronal excitability and propensity for seizure generation. In addition to this, neuroinflammation can also occur in AE because of Fcγ-dependent immune system activation. CASPR2 antibodies, for example, are able to activate both natural killer cells and macrophages through Fcγ receptor binding, likely contributing to the inflammatory environment.²⁵

To the best of our knowledge, our study provides the first demonstration that the mediators released by the secretome of AE-derived PBMCs, without any *in vitro* stimulation, are able to alter neuronal excitability and induce SLEs in the limbic area of an *in vitro* mammalian brain preparation. These findings suggest that peripheral immune cells of people with AE maintain an active profile and could contribute to the development of seizures in our patients.

Accordingly, Multiplex ELLA immunoassay analysis showed that the cytokine expression pattern is different between AE patients and healthy controls. In particular,

significantly lower concentrations were found of IL-10, IL-1Ra, TNF-α, IL-2, and IL-6 in the secretome of PBMCs derived from AE patients compared to healthy subjects, whereas IL-1β levels were comparable in both cohorts.

Our findings might appear to contrast with other studies that measured cytokine levels in both serum and CSF of patients with AE and drug-resistant epilepsy. Elevated concentrations of IL-6, IL-1β, TNF-α, interferon-γ, IL-17, IL-2, C-X-C motif chemokine ligand 10 (CXCL10), and CXCL13 have been reported in the CSF during the acute phase of AE,⁷ with IL-2 levels also increased in the serum. Similarly, in drug-resistant epilepsy, proinflammatory cytokines such as IL-1β, IL-6, and TNF-α are often present at high levels in both serum and CSF. Notably, serum IL-1β and IL-6 levels decline following surgical intervention and are associated with seizure reduction,²⁶ suggesting that seizures may drive the inflammatory abnormalities observed in these patients. In our experimental setting, the net effect of peripheral inflammation on neuronal network excitability may likely depend on the relative proportions and dynamic balance between pro- and anti-inflammatory cytokines, and not only on the absolute levels of each individual mediator.

We chose to focus on PBMCs because they represent an easily accessible source of immune cells providing an

intact source of biomarkers.²⁷ Moreover, cytokine production by PBMCs *in vitro* is more stable and less subject to the acute fluctuations seen in serum. Serum cytokine levels can be highly variable due to rapid turnover, binding to inhibitors, and acute systemic responses. PBMC secretome represents a mixed, cross-sectional sample of cytokines and provides information on the intrinsic functionality of the patient's immune cells.²⁸

Previous studies have shown that immune mediators, after entering the brain parenchyma, contribute to inducing ion channel and other receptor modifications that are permissive of neuronal hyperexcitability or glial cells activation.¹⁷ Anti-inflammatory cytokines (like IL-10 and IL-1Ra) are crucial for regulating the inflammatory response. Low levels might impair the brain's ability to dampen inflammation, potentially making it more susceptible to seizures.^{9,29}

The impact of IL-10 on GABAergic neurotransmission has been recently evaluated in human epileptogenic gangliogliomas, where GABAergic currents are potentiated by IL-10 and are inhibited by blocking IL-10 signaling, suggesting that an insufficient anti-inflammatory immune response due to reduced IL-10 signaling/bioavailability could account for the chronic disease state in epilepsy.³⁰

IL-1ra is an endogenous competitive blocker of IL-1R1, and its specific action is to terminate or prevent the biological actions of endogenous IL-1 β .³¹ The activation of the prototypical proinflammatory axis formed by IL-1 β and its cognate receptor type 1 (IL-1R1) is involved in the generation and recurrence of experimental seizures *in vivo*³² and *in vitro*.⁹ IL-1 β reduces synaptically mediated GABA inhibition in area CA3 of the hippocampus and potentiates NMDAR function by enhancing NMDA-mediated Ca⁺⁺ influx by activating Src kinase-catalyzed N-methyl-D-aspartate receptor subunit NR2B phosphorylation.³³

Due to the intrinsic limitation of this preliminary study, our results are insufficient to elucidate the precise effect of each immune mediator. However, we may hypothesize that in the secretome of PBMCs derived from AE patients, lower levels of IL-1Ra may be unable to rapidly counteract the effects of elevated IL-1 β levels. The human recombinant IL-1 β receptor antagonist anakinra has been demonstrated to rapidly affect the duration and the recurrence of seizure activity in *in vivo*³⁴ and *in vitro*⁹ models. Of note, it is also successfully used in patients with new onset refractory status epilepticus and febrile infection-related epilepsy syndrome.³⁵

The effect of the secretome from hPBMCs derived from AE patients was also analyzed regarding glial cell activation. Our experiments might suggest that the secretome from hPBMCs derived from AE patients and the associated SLEs rapidly change microglial cells' morphological phenotype from a resting to an amoeboid activated state.^{36,37}

Whether acute microgliosis represents a consequence or a cause of seizure activity cannot be ascertained in our acute experimental conditions, but it likely reflects an inflammatory state of brain parenchyma induced either by the secretome from hPBMCs derived from AE patients or by the seizures induced by the secretome.

Our study has limitations. First, data were collected from a limited number of subjects with heterogeneous subtypes of AE, and in seronegative cases we cannot exclude the possibility that they possess unidentified NSAbs, mainly when a full second-level testing was not performed. Further research involving larger cohorts of patients with AE without seizures, patients with epilepsy of nonimmunological origin, and those with systemic autoimmune diseases without seizures are needed to validate and better interpret these findings. Second, we analyzed only a limited subset of inflammatory modulators; other cytokines or their combinations may also contribute to ictogenesis.

5 | CONCLUSIONS

Our results represent a preliminary step toward a better understanding of peripheral immunological mediators, and particularly PBMC secretome components like cytokines, as a potential common fingerprint of immune-mediated seizures. These findings could translate into clinical practice to guide the management of seizures in patients with AE by considering additional mechanism that go beyond the traditional antibody-mediated mechanisms.

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CONFLICT OF INTEREST STATEMENT

The authors have nothing to disclose. All the authors confirm that they have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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